INDUSTRIAL AUTOMATION

Patryk Choiński



Field Application Engineer

PLC / HMI / Motion Control / SCADA

patryk@choinski.net.pl +48 696 568 545

Portfolio ↗ GitHub ↗ LinkedIn ↗

https://patrykchoinski.github.io/Portfolio/

01 Professional profile

I specialise in industrial automation applications: PLC, HMI, SCADA, motion control and drives. I connect controller logic with machine operation, actuators and operator interfaces. I use IT tools for data integration, visualisation and test automation.

02 Professional experience

03.2020 - present

Field Application Engineer

Company in the industrial automation sector

Developing PLC, HMI and motion control applications for stone, packaging and woodworking machines, as well as other industrial processes. Integrating servo drives, variable frequency drives and pneumatic systems.

10.2018 - 03.2020

Technical Support Engineer

Company in the industrial automation sector

Technical support for PLC, HMI, SCADA, drives, motion controllers and Delta robots.

07.2017 - 08.2017

Engineering intern

Company in the automotive sector

Programming Mitsubishi PLCs and Fanuc robots, and working on electrical systems.

03 Technical skills

PLC programming
PLC · Structured Text

Servo drives and variable frequency drives
Servo drives · Variable frequency drives · Encoders

Temperature controllers
PID · Hysteresis control

Robotics and electrical engineering
Robots · Pneumatics · Electrical systems

Motion control
Positioning · Axis homing · Interpolation · Motion profiles

HMI operator panels
HMI · Manual / automatic modes · Machine parameters

SCADA and process data
SCADA · Recipes · Trends and reports

Industrial communication
EtherCAT · CANopen · EtherNet/IP · Modbus TCP · Modbus RTU · OPC UA

04 Selected applications and projects

Mould cooling channel cleaning and diagnostics

Company in the injection mould servicing sector

I programmed the PLC and HMI for a machine used to clean and diagnose injection mould cooling channels.

Plastic packaging production

Company in the plastics processing machinery sector

I programmed the PLC and HMI for a machine producing plastic packaging.

Packaging machine automation

Company in the packaging machinery sector

I programmed control systems for cardboard packaging production machines.

Process control, recipes and reporting

Company in the industrial mixing systems sector

I programmed a SCADA application for a mixing process, including historical data and reporting.

Automated window blind production

Company in the blind manufacturing machinery sector

I programmed the control system for an automated window blind production machine.

Stone processing machine control

Company in the stone processing machinery sector

I programmed the PLC, HMI, servo drive and variable frequency drive control for a stone processing machine.

Automated PLC application verification

Company in the industrial automation sector

I programmed automated build, deployment and testing workflows for PLC applications.

05 IT supporting industrial automation

Web visualisation and tools

JavaScript · Node.js · WebGL · HTML · CSS · SSE

Databases and integrations

SQL Server · PostgreSQL · SQL · SQLite

AI in engineering workflows

Claude · Codex · AI

Scripting and automated tests

Python · PowerShell · Git · GitHub Actions · CI/CD · JUnit

06 Education

2014 - 2018 **Automation and Robotics - engineering studies**
Warsaw University of Technology

2019 - 2020 **Computer Science - period of study listed in the CV**
Warsaw University of Technology

07 Languages and training

English: B2. Training: electrical safety up to 1 kV (SEP, 2019); machine safety, EN 60204-1 (2022); first aid (2023).

Training dates as listed in the CV. Company names are anonymised. Application details, materials and demonstrations are available in the portfolio.